03 · Price elasticity by region — random slopes & endogeneity

July 11, 2026

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The business decision. We set prices region by region and want each region's **price elasticity of demand** — the standard economics measure of how sensitive demand is to price. Concretely, elasticity is the **% change in quantity sold for a 1% change in price**; it's almost always negative (raise price -> sell less). An elasticity of -1.5 means "cut price 1%, sell 1.5% more." We need it per region because a tourist market and a price-sensitive suburb should not be priced the same.

1.0.1 Two traps this notebook is built around

1. **Small regions are noisy.** A region with 12 weeks of data will hand you a wild elasticity estimate that's mostly sampling noise. Acting on it is overfitting. The fix is **partial pooling** (below).
2. **Price is endogenous.** "Endogenous" means the treatment (price) is *correlated with the error term* — here because managers historically *cut prices when they expected soft demand.* So low prices coincide with low demand for reasons that have nothing to do with the price cut, which drags the naive elasticity estimate *toward zero* (makes demand look less price-sensitive than it is). No amount of clever averaging fixes this — it needs an **instrument** (notebook 11), which we preview at the end.

1.0.2 The tools, in plain terms

- **Log-log demand.** We model $\log(\text{quantity}) = \alpha + \beta \log(\text{price}) + \ldots$. The magic of taking logs on both sides is that the slope β **is** the elasticity directly (a constant %-for-% relationship), so there's one interpretable number per region.
- **Random slopes / partial pooling.** Instead of one elasticity for all regions (ignores real differences) or a separate free estimate per region (noisy), we assume each region's elasticity β_r is drawn from a shared distribution $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta)$. This **hierarchical** structure makes each region's estimate a compromise between its own noisy data and the fleet average — data-poor regions get pulled ("shrunk") harder toward the fleet, which is exactly the protection we want. μ_β is the typical elasticity; τ_β measures how much regions genuinely differ.

  Cookbook maps this to `pathmc` random slopes; here we fit the equivalent hierarchical model explicitly in PyMC so the shrinkage is fully visible. The identification logic is unchanged.

  On real data. Swap in your **own transaction panel** — one row per region per week with price, units sold, and controls (competitor price, promotions, seasonality). Public analogues are retail *scanner* datasets (e.g. the Dominick's Finer Foods store data). The endogeneity caveat is not academic: it's the single biggest reason naive "price tests" mislead in practice.

1.1 2 · Simulate a ground truth

A region x week panel with a **constant-elasticity (log-log) demand curve** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where β_r is region r's true price elasticity (planted, varying around a fleet mean ~ -1.4). Of the observed covariates Z, only **competitor price** is a genuine confounder — it moves both our price and demand; **season** and **trend** shift demand only, so they enter as *precision covariates* (they sharpen estimates), not confounders. We start with **no** endogeneity so we can show recovery, then switch it on in step 7.

The data-generating model — exactly what `dgp.price_panel` implements (defaults & seed in `src/cmp/dgp.py`). For region r, week t: season $s_t = 0.3\sin(2\pi t/26)$, trend $g_t = 0.01\,t$, competitor price $C_{rt} \sim \mathcal{N}(20,\ 2^2)$, and a demand shock $u_{rt} \sim \mathcal{N}(0,1)$:

$$\beta_r \sim \mathcal{N}(-1.4,\ 0.35^2) \qquad \text{(true regional elasticit}$$

$$\log P_{rt} = \log 15\ -\ 0.05\, C_{rt}/20\ +\ \kappa\, u_{rt}\ +\ \nu_{rt}, \qquad \nu_{rt} \sim \mathcal{N}(0,\ 0.05^2) \qquad \text{(price-setting)}$$

$$\log Q_{rt} = 2.5\ +\ \beta_r \log P_{rt}\ +\ 0.4\,\tfrac{C_{rt}-20}{20}\ +\ s_t\ +\ g_t\ +\ u_{rt}\ +\ \varepsilon_{rt}, \qquad \varepsilon_{rt} \sim \mathcal{N}(0,\ 0.08^2).$$

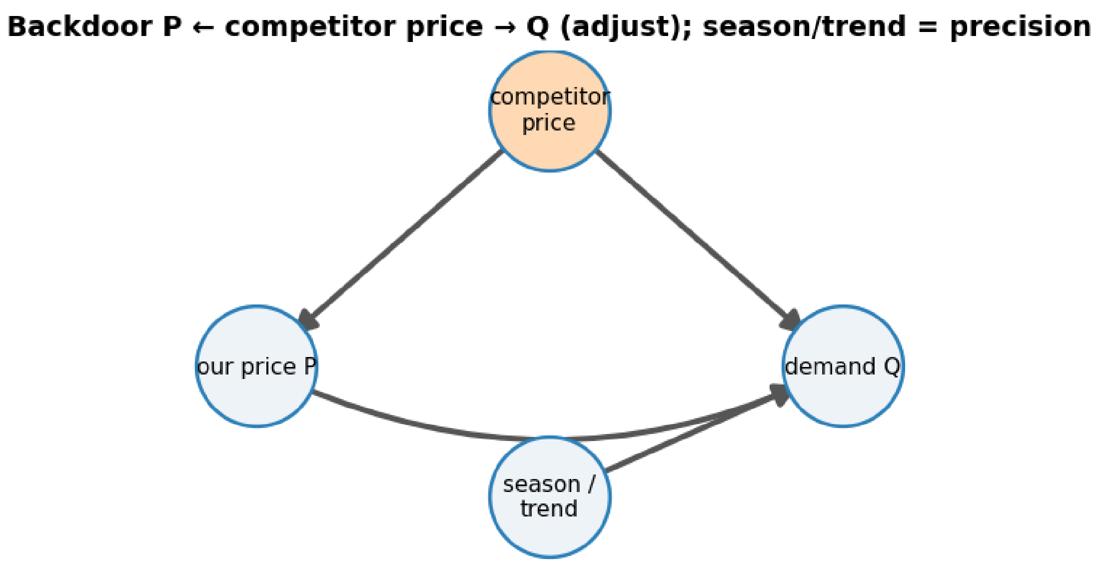
C_{rt} appears in **both** equations — that is what makes competitor price the genuine confounder (it moves our price *and* demand), while s_t and g_t appear only in the demand equation (precision covariates). κ is the **endogeneity switch**: through step 6 the generator runs with $\kappa = 0$ and zeroes the shock entirely ($u_{rt} \equiv 0$), so recovery is clean; step 7 sets $\kappa > 0$, and the *same* shock now enters both equations — managers' prices respond to the very shock that moves demand ($\mathrm{Cov}(\log P, u) > 0$), the simultaneity that biases the naive elasticity toward zero.

```
12 regions, fleet-mean elasticity -1.42; obs per region range 16-80 (regions 0-2
are thin, to exercise shrinkage)

      region  week      price    demand  competitor_price        season  \
0  region_00     2  13.003239  0.159630         20.405765  1.394170e-01
1  region_00     3  14.245949  0.157405         16.535730  1.989368e-01
2  region_00    12  14.310937  0.159097         20.819276  7.179470e-02
3  region_00    13  15.626376  0.175255         21.659711 -9.648736e-17
4  region_00    15  13.204126  0.152195         19.486540 -1.394170e-01

   trend  log_price  log_demand
0   0.02   2.565199   -1.834900
1   0.03   2.656473   -1.848934
2   0.12   2.661024   -1.838239
3   0.13   2.748960   -1.741513
4   0.15   2.580529   -1.882592
```

1.2 3 · Identify — elasticity as a random slope, and the backdoor set

Estimand. Per-region elasticity $\beta_r = \partial \log Q / \partial \log P$. **Partial pooling** treats $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta)$ — noisy regions shrink toward the fleet mean μ_β, and τ_β *measures how much elasticity genuinely* *varies*. The **shrinkage factor** for region r is approximately $1-\dfrac{\tau_\beta^2}{\tau_\beta^2+\sigma^2/n_r}$ — thin regions (small n_r) get pulled harder toward the fleet. Shrinkage is a feature: it stops us over-reacting to a small region's noisy slope.

Confounding is the crux. Price is endogenous — a demand shock moves both P and Q. The one true observable confounder is **competitor price** (it moves both P and Q); conditioning on it closes that backdoor path. **Season** and **trend** move demand only, so they are *precision covariates*, not confounders. Together this closes the *observable* paths. What pooling does **not** fix is an *unobserved* common shock — for that you need an instrument (notebook 11). We state this out loud rather than hide it.



1.3 4 · Estimate — three pooling regimes, side by side

To make the bias–variance trade-off concrete we estimate each region's elasticity **three ways**:

- **No pooling** — a separate regression per region. Unbiased, but a data-poor region's estimate is pure noise. This is the "trust each region's own numbers" extreme.
- **Complete pooling** — one elasticity for the whole fleet. Rock-steady, but pretends every region is identical, so it's biased for any region that genuinely differs. The "everyone's the same" extreme.
- **Partial pooling** — the hierarchical Bayesian middle ground: each β_r is nudged toward the fleet mean by an amount that depends on how much data the region has. Data-rich regions barely move; data-poor regions borrow heavily. This is the one that gets *both* the overall level and the spread right.

The model below reads directly as the maths from Step 3: `mu_beta` is the fleet-mean elasticity μ_β, `tau_beta` is how much regions differ τ_β, and `beta_r` are the per-region elasticities drawn from $\mathcal{N}(\mu_\beta, \tau_\beta)$. `alpha_r` are region baselines and `gamma` the controls (competitor price, seasonality, trend). The printout compares the recovered fleet mean to the planted truth.

The fitted model, in symbols (observation i in region $r[i]$; z_i the standardized controls — competitor price, season, trend):

$$
\begin{aligned}
\log Q_i &\sim \mathcal{N}\big(\alpha_{r[i]} + \beta_{r[i]} \log P_i + \gamma^\top z_i,\ \sigma^2\big)\\
\beta_r &\sim \mathcal{N}(\mu_\beta,\ \tau_\beta^2), \qquad \alpha_r \sim \mathcal{N}(0,\ 5^2), \qquad \gamma \sim \mathcal{N}(0,\ 2^2)\\
\mu_\beta &\sim \mathcal{N}(-1,\ 1^2), \qquad \tau_\beta \sim \mathrm{HalfNormal}(1), \qquad \sigma \sim \mathrm{HalfNormal}(1).
\end{aligned}
$$

The middle line's $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ **is** the partial pooling: τ_β (how much regions truly differ) is *learned from the data*, and each region's posterior elasticity is a precision-weighted compromise between its own weeks and the fleet mean — thin regions lean on the fleet, data-rich regions stand on their own.

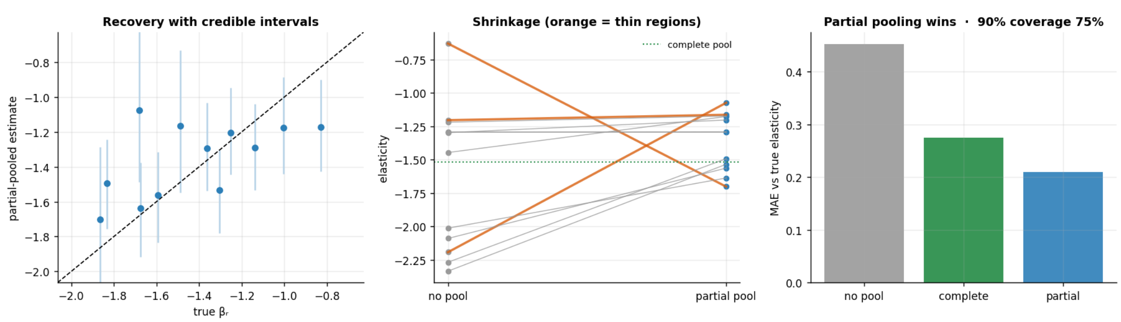
Like-for-like caveat. The no-pool and complete-pooling baselines are *bivariate* $\log Q$-on-$\log P$ fits (`np.polyfit`) — they omit the season/trend/competitor-price controls Z that the hierarchical model adjusts for, so their gap to partial pooling reflects **both** the pooling choice *and* the control choice; the hierarchical fit is the like-for-like causal estimate.

```
hierarchical model convergence: max r-hat 1.000 - min ESS 675 - divergences 0
fleet-mean elasticity mu_beta -1.36 (true -1.42)  ·  between-region spread
tau_beta 0.30 (true 0.31)
```

1.4 5 · Validate — recovery, shrinkage, and calibration

Estimated vs true elasticity should track the 45° line; the thin regions' point estimates get pulled toward the fleet mean (that's pooling protecting us), and the credible intervals are wider where data is thin. We also check that the 90% intervals actually **cover** the true elasticities (calibration).

```
MAE - no pool 0.45, complete 0.28, partial 0.21  ·  coverage 75%
```



How to read those three panels. *Left* — partial-pooled estimates track the true elasticities along the 45° line, and the error bars are visibly wider for the thin regions (less data -> more honest uncertainty). *Middle* — the shrinkage in action: lines connect each region's noisy no-pool estimate (left) to its partial-pooled estimate (right); the **thin regions (orange) get yanked hardest** toward the fleet, while data-rich regions barely budge. That selective pull is exactly the protection partial pooling buys. *Right* — the bottom line: mean absolute error against the truth is **lowest for partial pooling**, beating both extremes; no-pooling is noisy, complete-pooling is biased. (90% coverage runs below nominal here — 75%, i.e. 9 of 12 regions, in this sample —

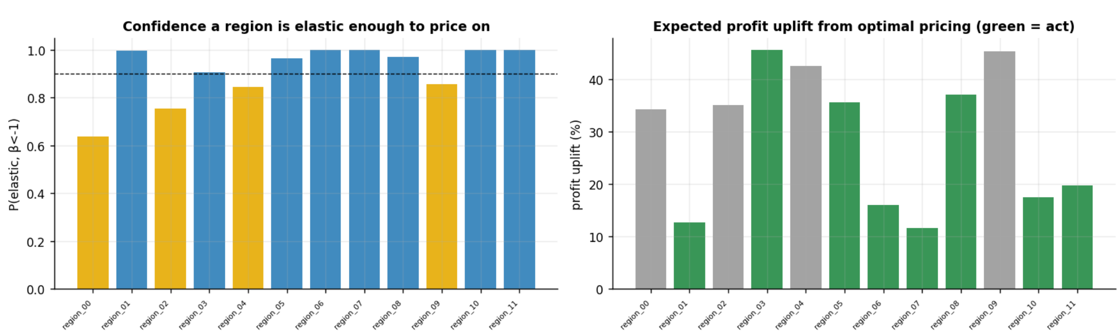
because the hard-shrunk thin regions have partial-pooling intervals that are slightly *too narrow* for their true, pre-shrinkage slope — a known small-sample property of hierarchical shrinkage intervals, not a bug; the point estimates are still recovered.)

1.5 6 · Decide, in euros — profit-maximising price per region

For a constant-elasticity demand curve with marginal cost c, the profit-maximising price is $P^{\star} = \dfrac{\beta}{\beta+1}c$ (needs $\beta < -1$). We propagate the **elasticity posterior** into a **price posterior** per region, and — the honest part — flag regions where we're not confident demand is even elastic ($\beta < -1$) for a **controlled price test** rather than acting on a shaky slope.

region	elasticity	P(elastic)	profit uplift %	P(uplift>0)	action
region_00	-1.07	0.64	34	0.64	controlled test
region_01	-1.70	1.00	13	1.00	set price
region_02	-1.16	0.75	35	0.75	controlled test
region_03	-1.20	0.91	46	0.90	set price
region_04	-1.18	0.84	43	0.84	controlled test
region_05	-1.29	0.97	36	0.97	set price
region_06	-1.56	1.00	16	1.00	set price
region_07	-1.64	1.00	12	1.00	set price
region_08	-1.29	0.97	37	0.97	set price
region_09	-1.17	0.86	45	0.86	controlled test
region_10	-1.53	1.00	17	1.00	set price
region_11	-1.49	1.00	20	1.00	set price

```
Across the 8 confidently-elastic regions, moving to optimal pricing lifts profit
by +24%
[90% +15%, +35%] on average - the pricing change's value, scale-free and with
uncertainty carried through (assumes constant elasticity out to P*; see
caveats).
```



How to read the decision table and the two panels. *Left* — each bar is $P(\beta < -1)$, our posterior confidence that a region's demand is **elastic** enough to price on: **blue** clears the 0.9 line (*set price*), **gold** sits in 0.5–0.9 (*controlled test*), **orange** is below 0.5 (*hold/test*). *Right* — the expected **profit uplift** from moving to the optimal price, **green** only where we are confident enough to act. The one fleet number underneath averages the uplift across the confidently-elastic regions and carries the posterior through to a 90% interval.

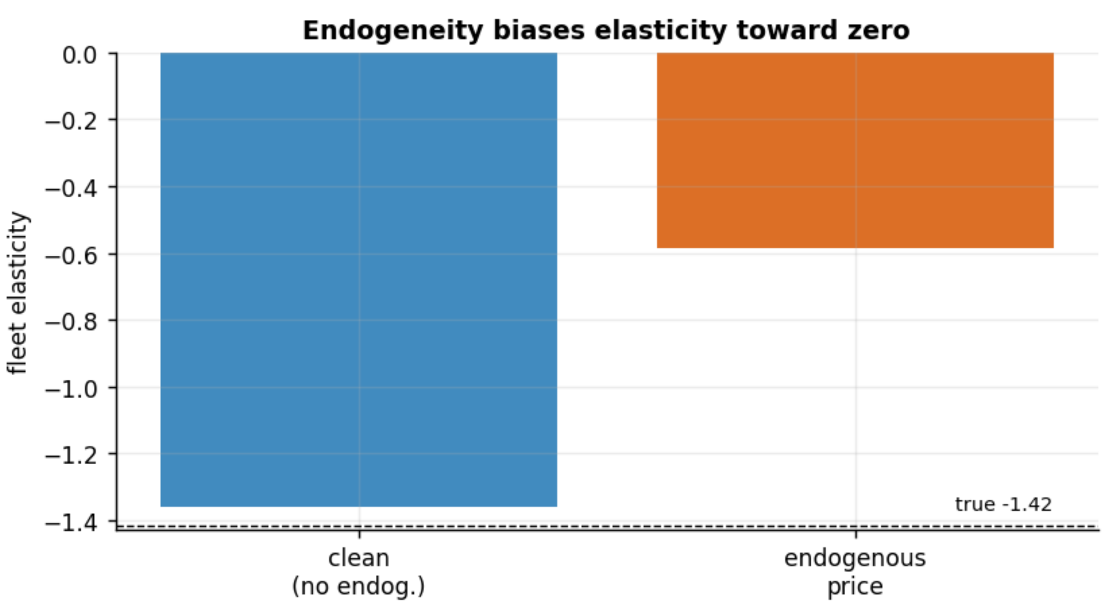
One subtlety worth saying out loud — the kind a sharp audience member will probe. The "profit uplift %" is a **posterior mean over *every* draw**, with the non-elastic draws ($\beta \geq -1$, where raising price doesn't pay) contributing **0%**. So it is an *expected* uplift already discounted by our uncertainty — not "the uplift once we act". For the confident **set price** rows ($P(\text{elastic}) \approx 1.00$) the two coincide; for a **controlled test** row like `region_00` ($P(\text{elastic}) = 0.64$) the printed **34%** is diluted by the 36% of draws worth 0% — the uplift *conditional on the region truly being elastic* is closer to $34/0.64 \approx$ **53**%.

This is also why you **cannot reproduce the column by plugging the row's own elasticity into the formula.** The closed form $P^\star = c\,\beta/(\beta+1)$ blows up as $\beta \to -1^-$: at `region_00`'s mean $\beta = -1.07$ it already implies $P^\star = 8 \cdot (-1.07)/(-0.07) \approx 122$ — an absurd **€122** next to the **€19** optima of the well-elastic regions. So the per-draw uplift has a heavy right tail and its **mean is not the formula evaluated at the mean elasticity** (Jensen's inequality, made severe by the boundary). We deliberately propagate the *whole* posterior rather than point-plug a single slope — and that very instability near $\beta = -1$ is the statistical reason those low-confidence regions are routed to a **controlled test**, not to a price move we might regret.

1.6 7 · Caveats — what partial pooling does *not* fix

Partial pooling handles *noise*, not *bias*. If price is endogenous to an **unobserved** demand shock, adjusting for observed Z isn't enough and elasticity is biased **toward zero**. We demonstrate by switching the hidden shock on — and preview the IV cure. (That re-run refits on the **full** panel, not the thinned subset used above — which only sharpens the fleet estimate and doesn't change the bias point being made.)

```
clean fleet elasticity -1.36 (true -1.42)
endogenous fleet elasticity -0.59 (true -1.42) -> biased toward zero by +0.83
Fix = an instrument (a cost shifter) or a natural price experiment - see
notebook 11 (IV).
endogenous-fit convergence: max r-hat 1.000 - min ESS 5640 - divergences 0
```



Other caveats. Log-log assumes *constant* elasticity within region (a local approximation if elasticity varies with price level); `lag(price)` would capture stockpiling dynamics (omitted for clarity); and shrinkage is a prior choice — too few regions or a mis-specified τ_β prior can over-pool, so always inspect the fleet-vs-region spread.